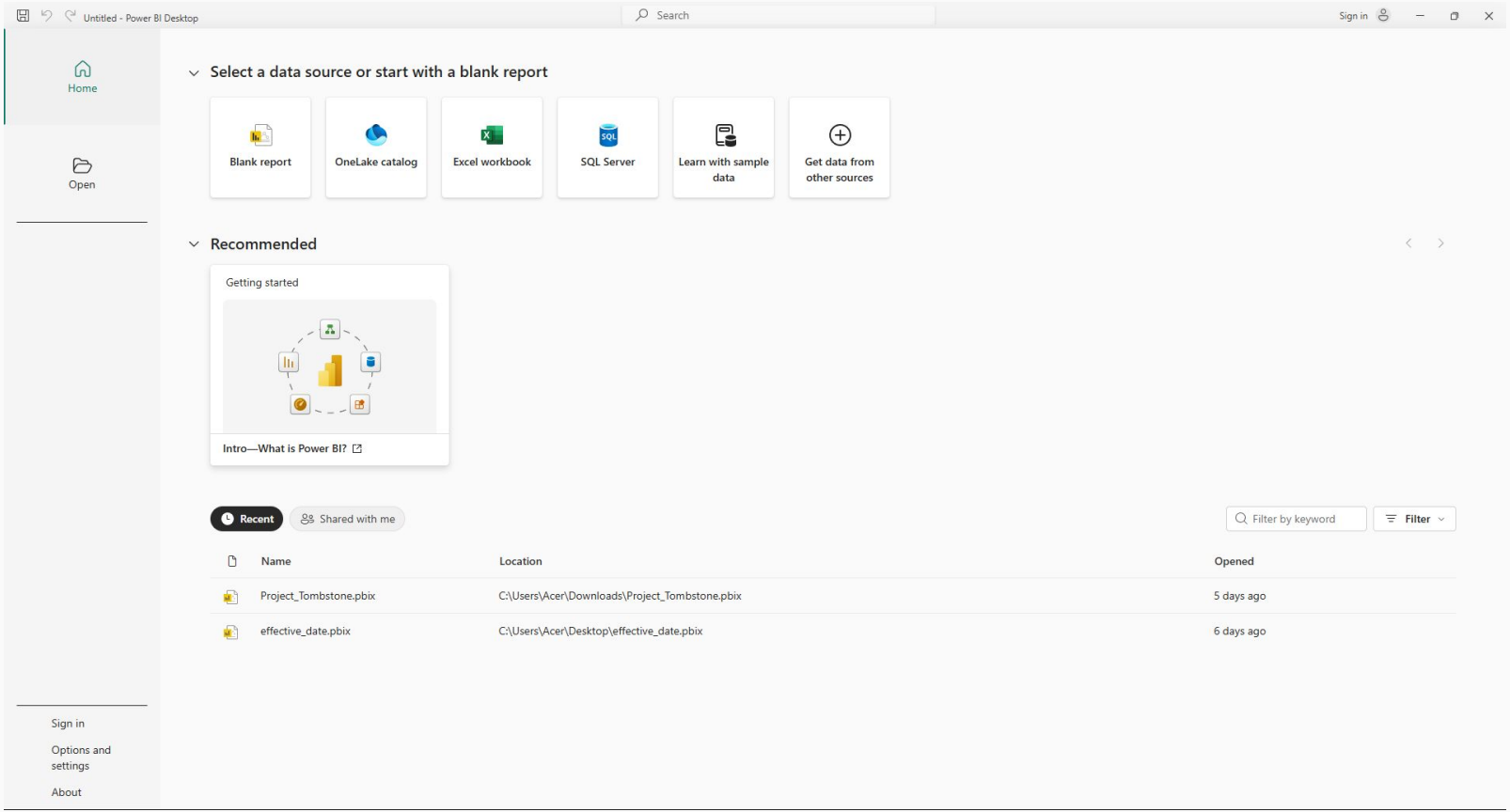
Introduction to Business Intelligence & Power BI

Kenneth A. Camasura

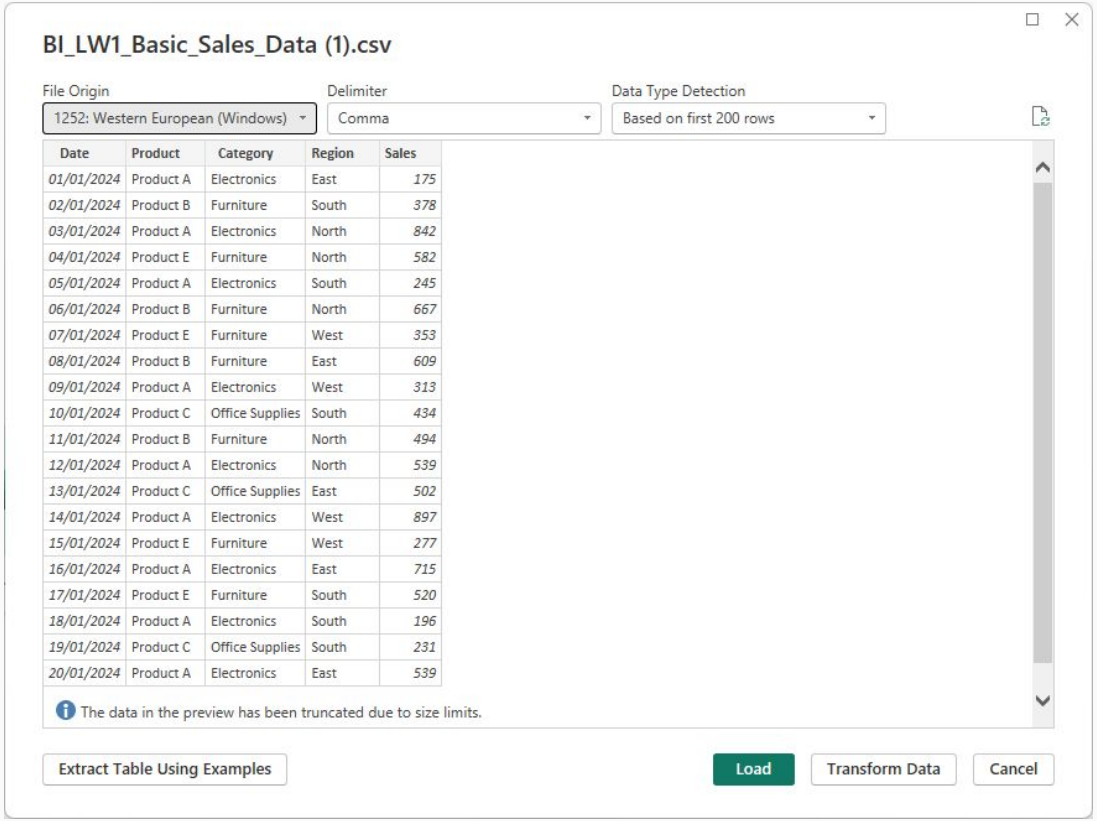
Instructor: Joseph A. Vistal

IS 107 - Business Intelligence

Part 1 | Step 1 : Open Power BI Desktop

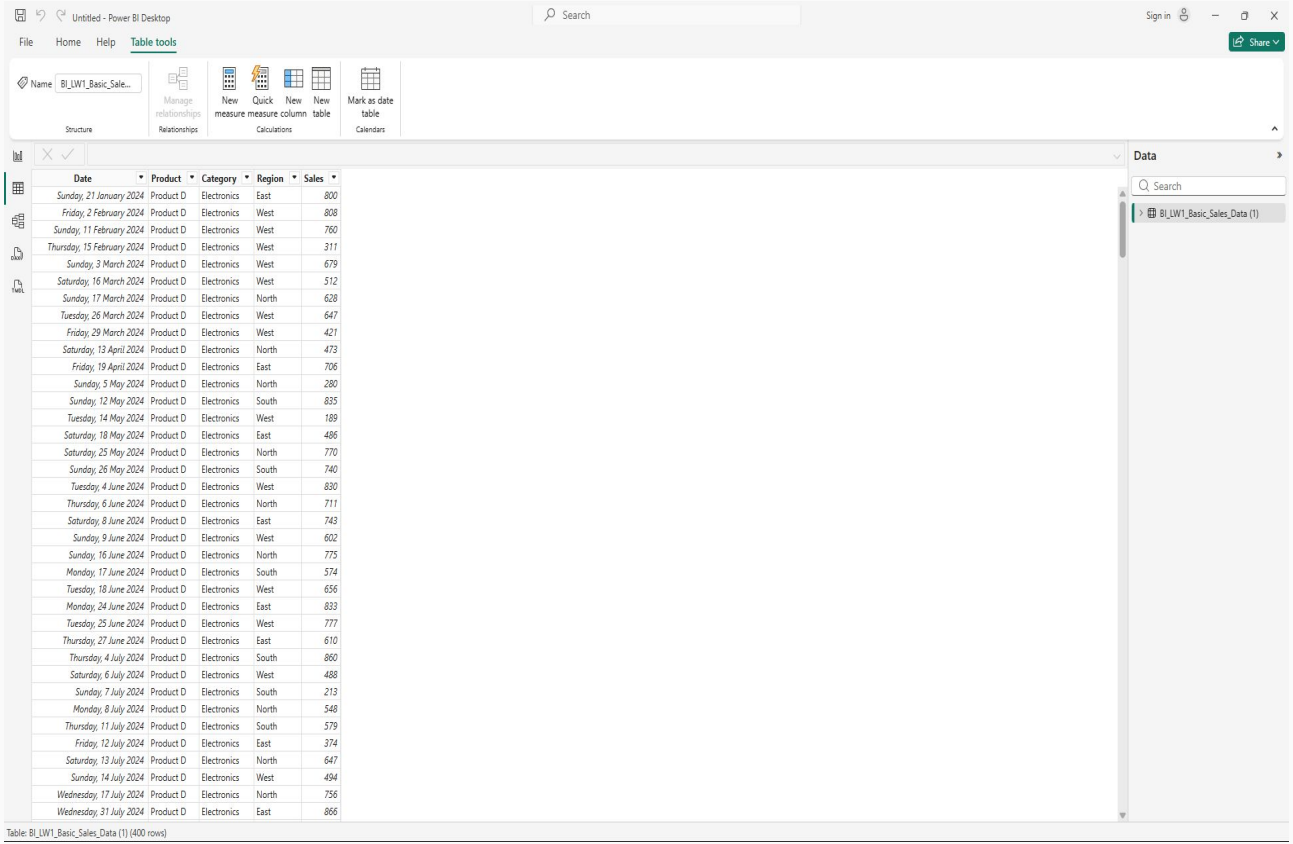


Part 1 | Step 2: Load the Dataset



Date	Product	Category	Region	Sales
01/01/2024	Product A	Electronics	East	175
02/01/2024	Product B	Furniture	South	378
03/01/2024	Product A	Electronics	North	842
04/01/2024	Product E	Furniture	North	582
05/01/2024	Product A	Electronics	South	245
06/01/2024	Product B	Furniture	North	667
07/01/2024	Product E	Furniture	West	353
08/01/2024	Product B	Furniture	East	609
09/01/2024	Product A	Electronics	West	313
10/01/2024	Product C	Office Supplies	South	434
11/01/2024	Product B	Furniture	North	494
12/01/2024	Product A	Electronics	North	539
13/01/2024	Product C	Office Supplies	East	502
14/01/2024	Product A	Electronics	West	897
15/01/2024	Product E	Furniture	West	277
16/01/2024	Product A	Electronics	East	715
17/01/2024	Product E	Furniture	South	520
18/01/2024	Product A	Electronics	South	196
19/01/2024	Product C	Office Supplies	South	231
20/01/2024	Product A	Electronics	East	539

Part 1| Step 3 Verify Data in Data View



Date	Product	Category	Region	Sales
Sunday, 21 January 2024	Product D	Electronics	East	800
Friday, 2 February 2024	Product D	Electronics	West	808
Sunday, 11 February 2024	Product D	Electronics	West	760
Thursday, 15 February 2024	Product D	Electronics	West	311
Sunday, 3 March 2024	Product D	Electronics	West	679
Saturday, 16 March 2024	Product D	Electronics	West	512
Sunday, 17 March 2024	Product D	Electronics	North	628
Tuesday, 26 March 2024	Product D	Electronics	West	647
Friday, 29 March 2024	Product D	Electronics	West	421
Saturday, 13 April 2024	Product D	Electronics	North	473
Friday, 19 April 2024	Product D	Electronics	East	706
Sunday, 5 May 2024	Product D	Electronics	North	280
Sunday, 12 May 2024	Product D	Electronics	South	835
Tuesday, 14 May 2024	Product D	Electronics	West	189
Saturday, 18 May 2024	Product D	Electronics	East	486
Saturday, 25 May 2024	Product D	Electronics	North	770
Sunday, 26 May 2024	Product D	Electronics	South	740
Tuesday, 4 June 2024	Product D	Electronics	West	830
Thursday, 6 June 2024	Product D	Electronics	North	711
Saturday, 8 June 2024	Product D	Electronics	East	743
Sunday, 9 June 2024	Product D	Electronics	West	602
Sunday, 16 June 2024	Product D	Electronics	North	775
Monday, 17 June 2024	Product D	Electronics	South	574
Tuesday, 18 June 2024	Product D	Electronics	West	656
Monday, 24 June 2024	Product D	Electronics	East	833
Tuesday, 25 June 2024	Product D	Electronics	West	777
Thursday, 27 June 2024	Product D	Electronics	East	610
Thursday, 4 July 2024	Product D	Electronics	South	860
Saturday, 6 July 2024	Product D	Electronics	West	488
Sunday, 7 July 2024	Product D	Electronics	South	213
Monday, 8 July 2024	Product D	Electronics	North	548
Thursday, 11 July 2024	Product D	Electronics	South	579
Friday, 12 July 2024	Product D	Electronics	East	374
Saturday, 13 July 2024	Product D	Electronics	North	647
Sunday, 14 July 2024	Product D	Electronics	West	494
Wednesday, 17 July 2024	Product D	Electronics	North	756
Wednesday, 31 July 2024	Product D	Electronics	East	866

Are all columns visible?

Answer: Yes

Is “Date” formatted as Date?

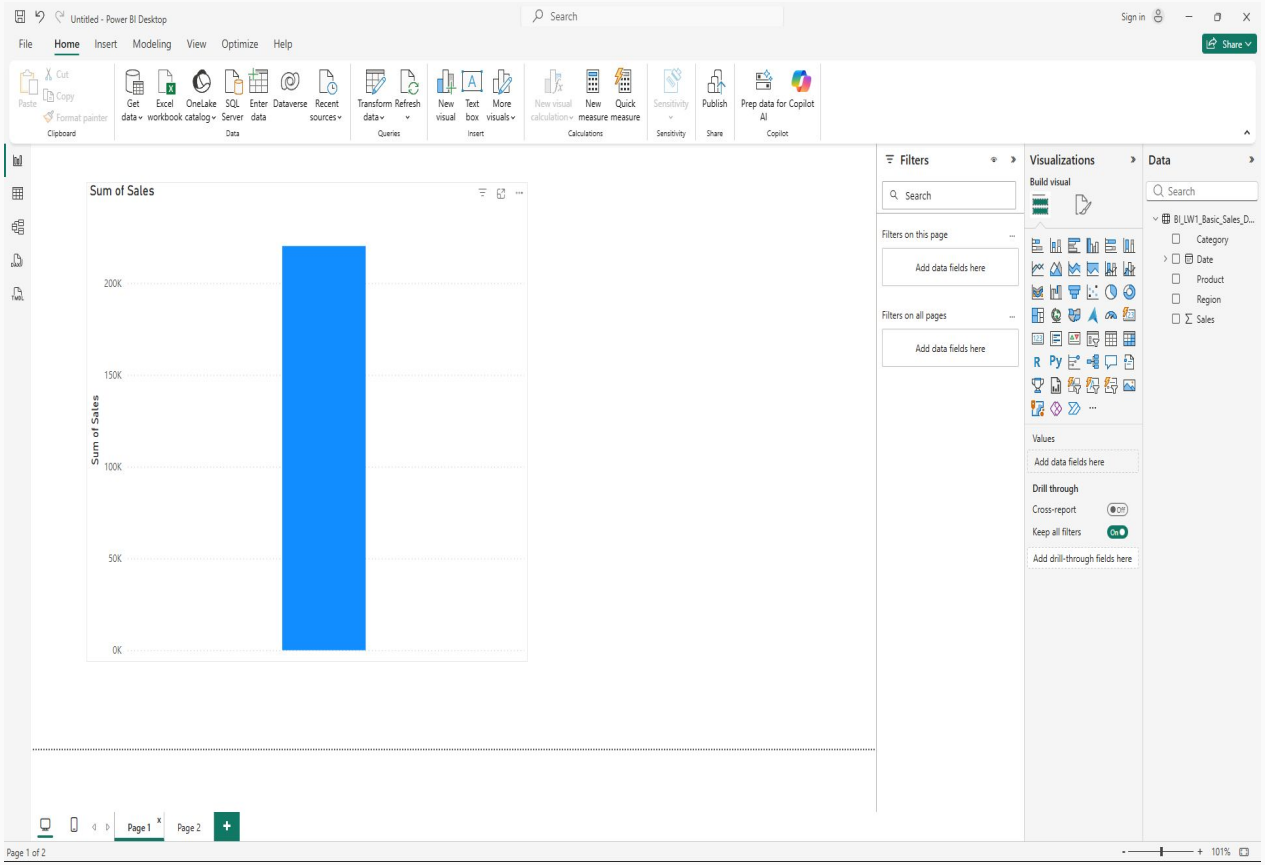
Answer: Yes

Is “Sales” formatted as Decimal Number?

Answer: No

Creating Auto-Generated Visuals

Part 3 | Step 1 Quick Visualization



What type of chart was created?

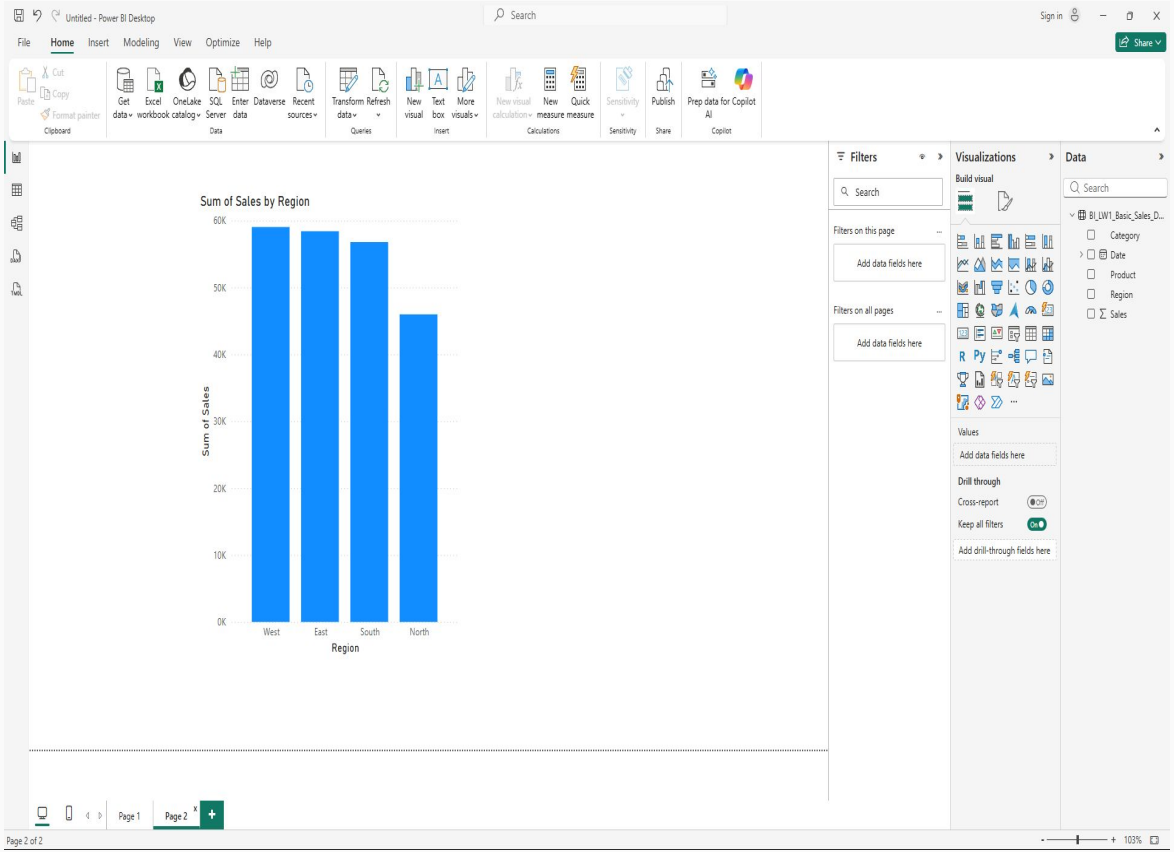
Answer: 100% Stacked Column Chart

What does it show?

Answer: Sum of Sales

Creating Auto-Generated Visuals

Part 3 | Step 2 Create a Sales by Region Chart

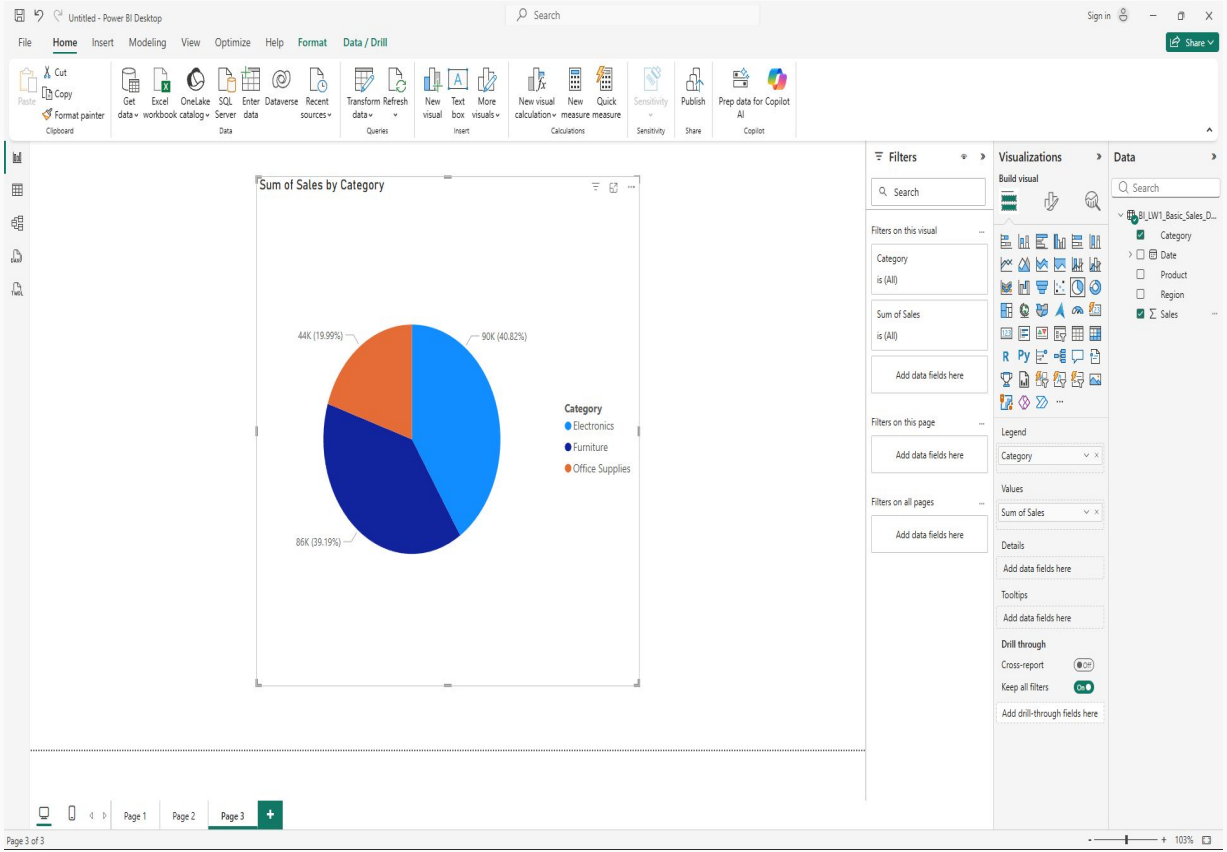


Which region has highest sales?

Answer: West Region

Creating Auto-Generated Visuals

Part 3 | Step 3 Sales by Category



Which category dominates?

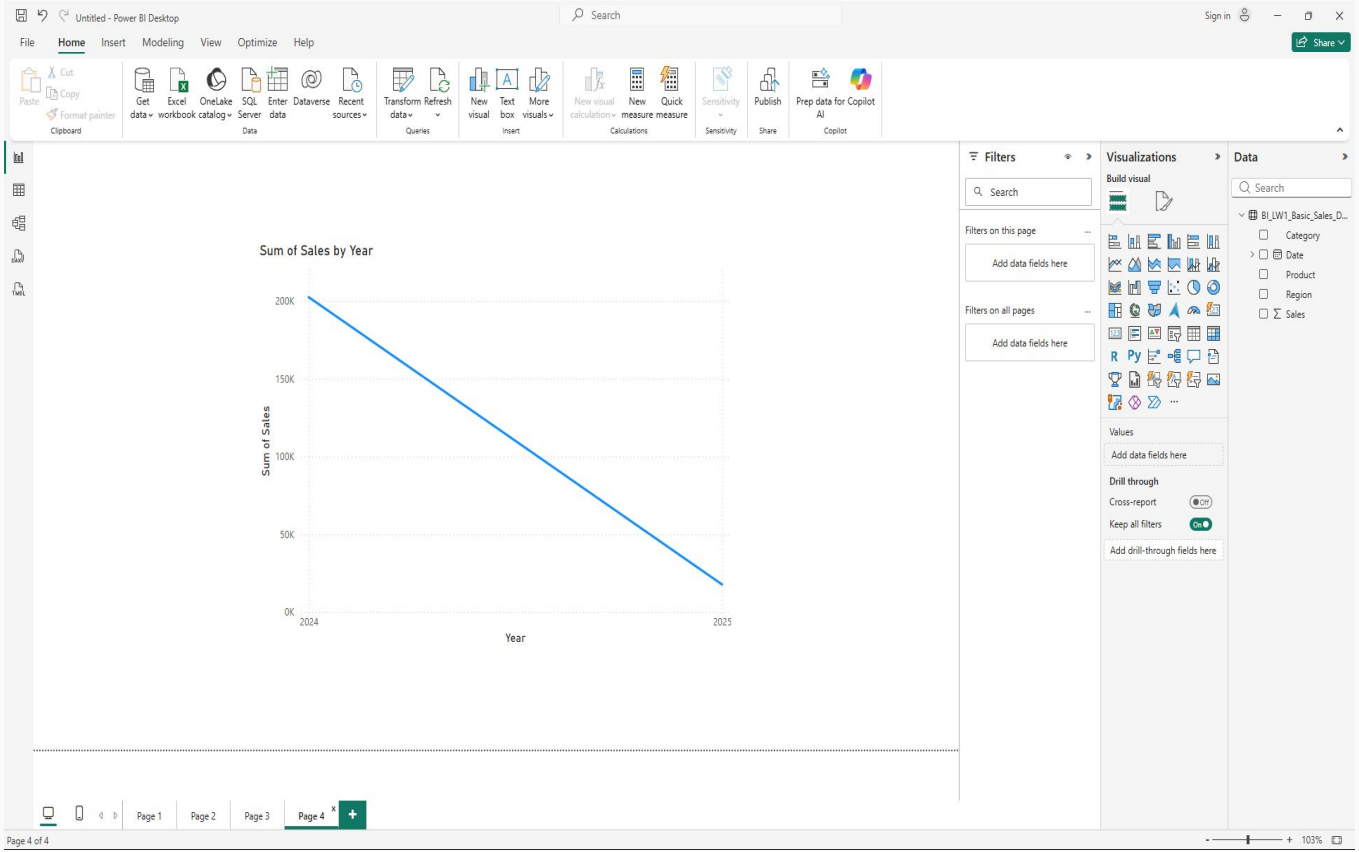
Answer: Electronics

Is the distribution balanced?

Answer: No

Creating Auto-Generated Visuals

Part 3 | Step 4 Sales by Category



Is there growth?

Answer: None

Any noticeable trend?

Answer: None

Basic Data Insight Interpretation

Which region contributes most revenue?

Answer: West

Which product category performs best?

Answer: Electronics

Are sales consistent across dates?

Answer: No

What business recommendation can you suggest?

Answer: Investing in North region.

Laboratory Questions

Part A – Technical Questions

What are the five columns in the dataset?

Answer: Date, Product, Category, Region and Sales

What data type is assigned to the “Sales” column?

Answer: Whole Number

Which Power BI view allows you to see raw data?

Answer: Table View

What chart type is best for showing trends over time?

Answer: Line Chart

What aggregation is automatically applied to Sales?

Answer: SUM

Laboratory Questions

Part B – Technical Questions

Which region has the highest total sales?

Answer: West

Which category has the lowest performance?

Answer: Office Supplies

Are sales increasing, decreasing, or stable?

Answer: Decreasing

If you were a manager, which region would you prioritize?

Answer: North

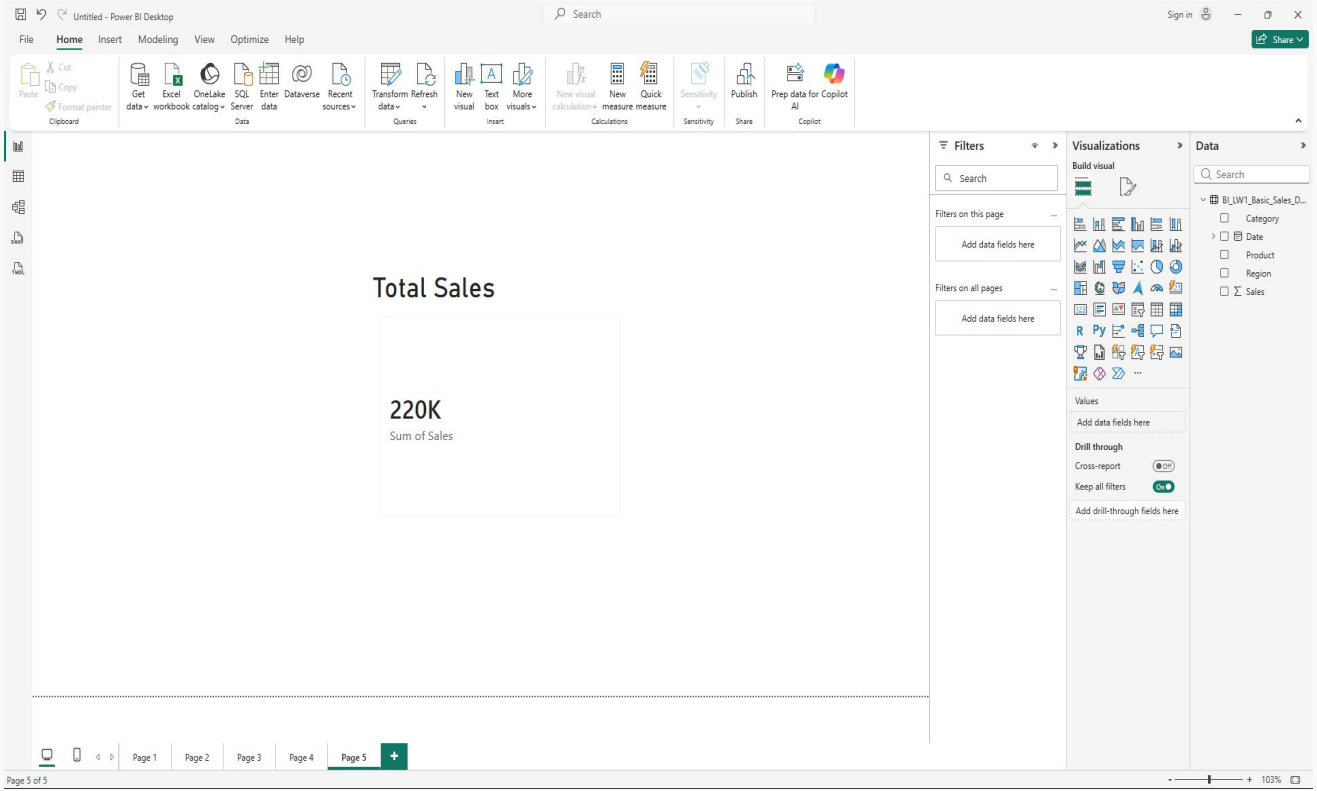
Provide one actionable recommendation based on the data.

Answer: Launch a targeted Office Supplies marketing campaign in the North region.

ENHANCEMENT SECTION

Advanced Exploration

Task 1 Add a Card Visualization



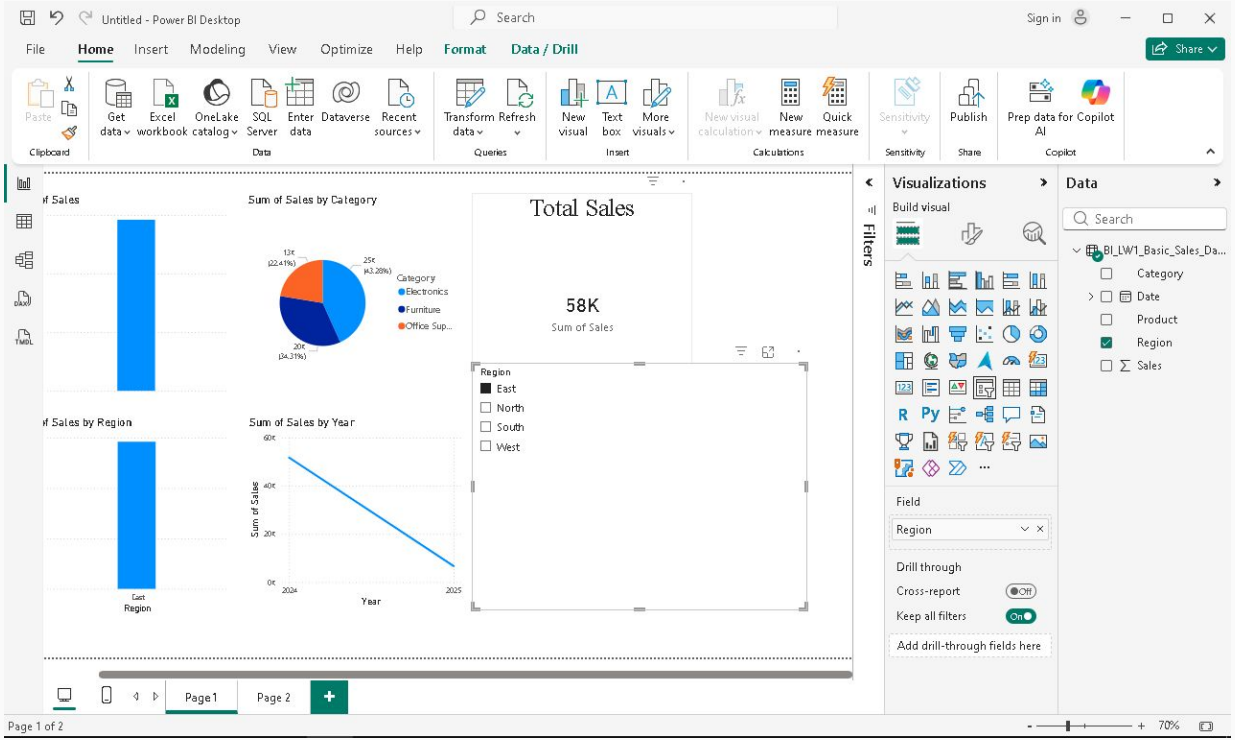
What is the total sales amount?

Answer:220k

ENHANCEMENT SECTION

Advanced Exploration

Task 2 Add Slicer



What happens to other visuals when you click a region?

Answer: The other visual will change.

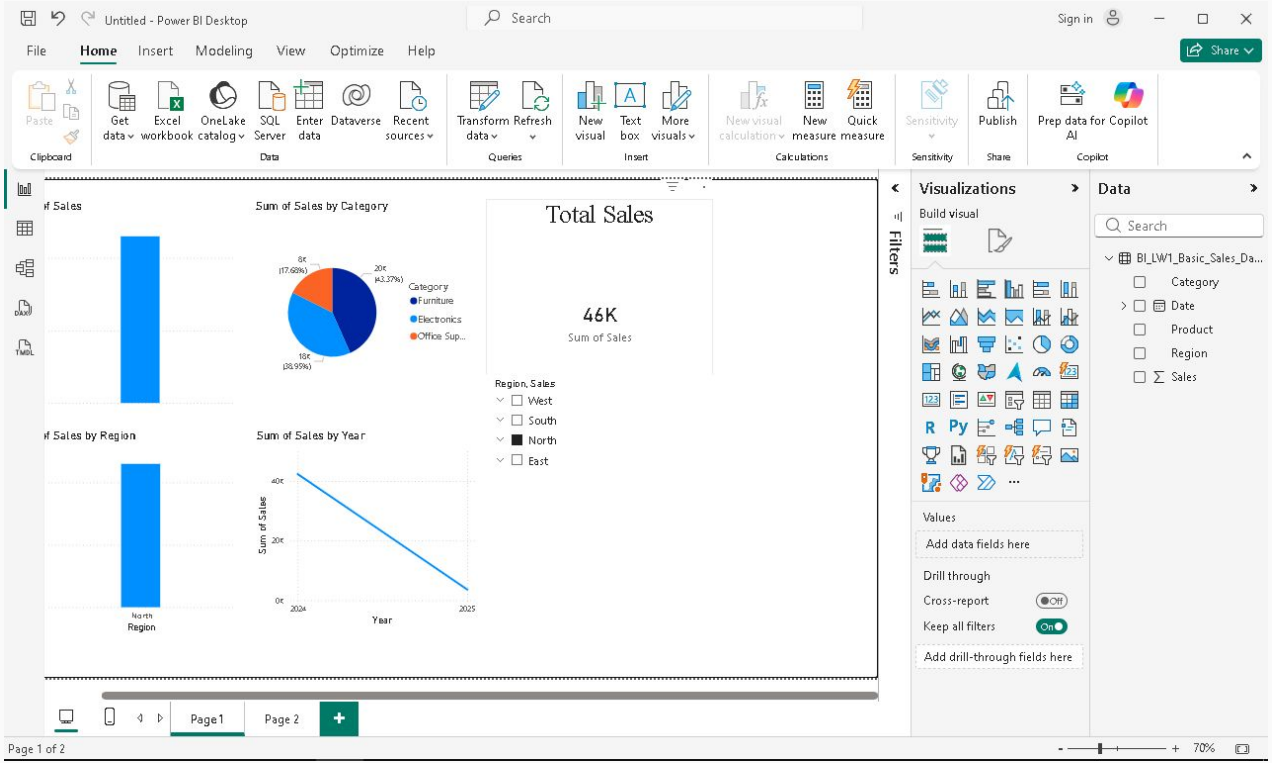
Why is filtering important in BI?

Answer: Filtering is important in BI because it transforms overwhelming data into actionable insights.

ENHANCEMENT SECTION

Advanced Exploration

Task 3 Sort Sales



Does sorting improve readability?

Answer: No

Why?

Answer: The sorting only rearrange the region in the card in descending order but the value still the same.

ENHANCEMENT SECTION

Advanced Exploration

Task 4 Identify Outliers

Which region is significantly higher or lower?

answer: West region is higher and North region is lower.

What might explain that difference?

answer: The sales has a big impact between this two region because it proves that west region has a lot more buyers than in the north region.